

WILDLIFE BIOLOGY

Research article

Demographic analysis of behavioral reactions of African bush elephants to a synthetic honey bee alarm pheromone blend

Mark G. Wright¹  ¹, Michelle G. Au¹ , Irene Gatti^{1,2}, Juliana Salehi¹, Craig R. Spencer² , Paul Allin²  and Agenor Mafra-Neto³ 

¹Department of Plant and Environmental Protection Sciences, University of Hawai'i at Mānoa, HI, USA

²Transfrontier Africa, OWNRR, Hoedspruit, South Africa

³ISCA Inc., Riverside, CA, USA

Correspondence: Mark G. Wright (markwrig@hawaii.edu)

Wildlife Biology

2025: e01419

doi: [10.1002/wlb3.01419](https://doi.org/10.1002/wlb3.01419)

Subject Editor: Ezequiel Fabiano

Editor-in-Chief: Ilse Storch

Accepted 23 January 2025



Human–elephant conflict situations continue to be a growing conservation issue. Conflict situations typically arise as the result of elephants traversing places with infrastructure where they become disruptive to human activities. Developing passive options for managing elephant movement and reducing human–elephant conflict situations is a conservation priority. In this paper, we report elephant responses to a synthetic honey bee alarm pheromone blend, and the relationship between strength and frequency of deterrence responses from elephants. We show that a blend of isoamyl acetate, 2-heptanol, and 2-nonanol, volatile chemicals known to stimulate honey bee vigilance, defence, and mass attack, produced stronger repellence than isoamyl acetate alone, or blended with 2-heptanol. Subadult elephants, irrespective of sex, were the demographic group most likely to show deterrence responses to the alarm blends. In some cases, entire herds of elephants were repelled from alarm pheromone treatments placed at waterholes, suggesting that the method has potential for manipulation of elephant movements on a landscape. Integration of synthetic honey bee alarm pheromones with other elephant behavioral management tools is discussed.

Keywords: Alarm pheromone, honey bee, human–elephant conflict

Introduction

Ever-reducing space for conservation land and wild areas, with proximal human populations, causes increased frequency of human–wildlife conflicts (Ngama et al. 2016, Cook et al. 2017, Angelici et al. 2023, Ma et al. 2024). In Africa and Asia, elephants are frequently involved in human–elephant conflict, including damage to subsistence and commercial crops, disrupting traffic by wandering onto roads, damaging water supply pipes, impacting large trees in conservation areas, and posing risks of injury and even death to farmers attempting to protect their crops. These conflict situations typically result in negative outcomes for the elephants, such as stressful deterrence or removal, or



www.wildlifebiology.org

© 2025 The Author(s). Wildlife Biology published by John Wiley & Sons Ltd on behalf of Nordic Society Oikos

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

shooting of problem animals. Tools for containing elephants within conservation areas, or managing their behavior inside and outside of them have been developed or are being investigated, including physical boundaries (e.g. electric fences) and a diversity of potential deterrents, such as noxious-smelling plant extracts, sounds (e.g. loud percussive noises, recorded agitated honey bee sounds), honey bee hive fences, synthetic honey bee alarm pheromone blends, and defensive chemicals from ants (King et al. 2017, King et al. 2018, Wright et al. 2018, 2023, Tiller et al. 2022, Cook et al. 2023). Each of these management options contributes to managing elephant issues; however, most are ineffective when used in isolation. The development of a suite of management options that may be used in an integrated manner will likely provide the most effective and sustainable approach to manage elephants. Elephants have highly sensitive olfactory physiology with substantially larger numbers of olfactory receptors than most mammals (Greenwood et al. 2005, Plotnik et al. 2014), and the use of odors as warning cues may contribute to the suite of elephant behavioral management options.

There are several examples of the effectiveness of olfactory cues influencing elephant behavior. Work in Kenya has shown that elephants are very effectively deterred from feeding on trees inhabited by aggressive ant colonies (*Crematogaster* spp., Formicidae) that live in the galls they produce on *Acacia* trees (Goheen and Palmer 2010, Palmer and Brody 2013). When disturbed by elephants and other herbivores, the ants exit the galls in large numbers, and release alarm pheromones, volatile chemical signals (Wood et al. 2002), which accompany their attacks on the herbivore. Elephants are likely deterred by the ant's alarm pheromones. Elephants are also deterred by honey bees (*Apis mellifera*, Apidae), which sting them around their eyes, ears, and trunk. Sounds of bees have been shown to cause elephants to be displaced greater distances than other acoustic stimuli such as human-generated noise (Mortimer et al. 2021). Honey bee hive fences have been shown to effectively protect farms in Kenya (King et al. 2017) and trees in South Africa (Cook et al. 2017). When guard bees perceive threats, they release alarm pheromones which result in the honey bee colony initiating a massive attack on the source of disruption. We have previously established that elephants exhibit increased distress behavior, and/or are deterred by the presence of a synthetic blend of a subset of compounds of the bee alarm pheromone blend (Wright et al. 2018). The responses reported by Wright et al. (2018) to the blend containing isoamyl (isopentyl) acetate and heptanone were generally distinct but mild; the responses were typically signs of uncertainty displayed by the elephants, and changes in walking direction. The alarm pheromone blend of *A. mellifera* comprises at least 23 volatile compounds and possibly as many as 40, each contributing in different ways to the extent of the defensive response in the bees (Nouvian et al. 2016). Insect pests are often managed using semiochemicals and it is often the case that specific combinations of pheromone chemical components and their relative doses determine the effectiveness of the pheromone blend used (Howse et al. 1998, Witzgall et al. 2010), and considering

the highly sophisticated olfactory capabilities of elephants, their behavior may be equally influenced by the blends of volatile compounds they perceive. Our previous work with synthetic alarm pheromone blends to elicit elephant deterrence included only two compounds from the known assemblage of volatile chemicals that comprise the honey bee alarm pheromone complex. These chemicals were selected based on their demonstrated effect in mediating bee responses to threats (Novian et al. 2016), and assuming that the olfactory cues of this incomplete blend were likely to be perceived as a signal of danger to an observer with acute olfactory sensitivity. The current work examines the effect of a three component blend (isoamyl acetate, 2-heptanone, and 2-nonanol), known to elicit the strongest defensive response in *A. mellifera* (Nouvian et al. 2016), on elephant behavior. Isoamyl acetate was the first discovered component of the bee alarm pheromone complex; it attracts and alerts guard bees (Boch et al. 1962), and they have been shown to attack and sting a moving object once alerted (Ghent and Gary 1962). Isoamyl acetate is also the main component of the alarm pheromone blend (Allan et al. 1987). The function of 2-heptanone in the alarm blend is somewhat controversial, but it has been shown to frequently induce a vigorous response in guard bees (Collins and Blum 1982, reviewed by Novian et al. 2016). Responses of bees were statistically similar for 2-heptanol and 1-hexanol (Collins and Blum 1983). Collins and Blum (1982) showed that 2-nonanol also stimulated a strong, rapid alarm response in honey bees. Novian et al. (2016) rank 2-heptanone, 1-hexanol, and 2-nonanol as the three compounds producing the strongest overall alarm response in honey bees, with isoamyl acetate the major component by volume (5-10-fold produced compared to other compounds; Allan et al. 1987) of the alarm pheromone blend, and rank it within the top five compounds in terms of strength of honey bee response.

Despite the potentially profound survival outcomes of animal behavioral responses, intraspecific variation (such as demographic factors) in responses to various stimuli is seldom analyzed. In fact, less than 10% of 589 published papers examine this aspect. However, in the few studies that do, more than 75% report significant effects (Harding et al. 2019). Elephant behaviors vary with age and sex demographics, habitat and resource availability, and herd size, resulting from instinctive fixed action patterns, learned responses to stimuli, and interactions with herd members of different ages and social hierarchies (Mumby and Plotnik 2018). For example, young adult African elephant bulls in small groups are typically more likely to breach fences (although some cows do indeed breach fences, Mutinda et al. 2014), whereas herd matriarchs may be more likely to exhibit behaviors benefitting the entire herd. Additionally, other herd members may issue alarm warnings that trigger responses from multiple members of the herd (Soltis et al. 2014). Elephant behavior can be influenced by experience, including anthropogenic disturbances, with different age groups and sexes exhibiting different responses (Kioko et al. 2013). Human impacts can significantly affect elephant populations (Wittmeyer et al. 2021).

It is possible that responses to *A. mellifera* alarm pheromones may vary by demographic status of elephants perceiving the cues. This paper describes the type and magnitude of responses of elephants and their demographic variables (sex, age, herd size), to synthetic *A. mellifera* alarm pheromone blends. We discuss the potential for incorporating synthetic bee alarm pheromone blends with other intervention options for managing human–elephant conflict situations.

Material and methods

This work was conducted in the Olifants West Nature Reserve (OWNR, Limpopo Province, South Africa, 24°13'S 30°52'W), a private nature reserve, and part of the Greater Kruger Park, South Africa, and Jejane Private Nature Reserve, approximately 30 km south of OWRN. OWRN has large numbers of elephants at certain times of year, and large transient herds may frequently be observed feeding in the area and visiting water holes. Jejane also has transient herds, often the same ones in OWRN, that typically visit during the winter months. These are wide-ranging elephants that are able to move into the entire Greater Kruger Park, Kruger National Park, and into neighboring Mozambique.

To test whether a three-component blend of selected volatile compounds known to launch defensive responses in *A. mellifera* would result in greater repellence of elephants than the earlier tested two-component blend (Wright et al. 2018), we formulated isoamyl acetate, 2-heptanone, and 2-nonanol (20% V/V) with SPLAT[®] (Specialized Pheromone and Lure Application Technology, ISCA Technologies, Riverside, CA, USA) and deployed the three-component formulation at waterholes in the OWRN as previously described by Wright et al. (2018), to observe elephant reactions. Waterholes (1–3 per week depending on availability) with observation hides were selected for the work. They were large enough to permit deployment of at least a control and treatment at each location, and in some locations, two treatments simultaneously (Supporting information). Experiments were conducted weekly in 2019, then in 2021 to 2024. The three-component blend and untreated control were deployed at waterholes between 08:00 and 15:00, placing the three-component blend formulation into white socks suspended from branches around the waterholes. We avoided using the same branches for consecutive experiments. White socks were used to provide a noticeable but non-threatening means of attracting the elephant's attention to the contents. The experiments were conducted with at least one-week intervals between any

exposure at any specific location. The blends with the chemical cues were replaced at each exposure. Responses were video recorded by observers hidden within game-viewing hides, and analyzed to detect responses to the treatments. Elephants moving within 1 m of the treatments were considered to be responders. In cases where other individuals were apparently alerted by the responder, the herd response was considered a single response, rather than the sum of individuals within the herd. Responses were categorized on an ordinal scale (Wright et al. 2023) as detailed in Table 1.

The rankings (1–3) of responses we recognized are broadly comparable with those used by van de Water et al. (2020) for Asian elephants (relaxed; attentive; alarmed), and Goodyear and Shulte (2015) (vigilance; distress; avoidance). The responses of elephants to socks containing the three-component blend were compared with their responses to the two-component blend (IAA and 2-heptanol), reanalyzing the data from Wright et al. (2018), as detailed above. Emphasis was placed on differences in response intensity (as described above) and the sex and age of responding animals to better understand the effect of adding 2-nonanol to the blend.

Data analysis: Elephant sex and age demographics and herd size response data were analyzed in JMP[®] 17 Pro (SAS Institute Inc., Cary, NC, 1989–2021), using the ordinal logistic regression option to analyze the ranked responses. Probabilities of behavioral responses of different strengths in varying herd sizes, and by sex and age group, were modeled in the JMP[®] 17 Pro prediction profiler option. Overall probability of any response by treatment, age distribution, sex, and herd size were analyzed using the JMP generalized linear model procedure, with a binomial distribution and logit link function, followed by single degree of freedom contrast tests for effects. Probabilities of responding to the synthetic pheromone were analyzed using the GLM procedure with normal distribution and log link function for profile estimate comparisons; single degree of freedom contrasts were used to compare estimates for the age categories for substantial concern, and strong repellence (response levels 2 and 3, respectively).

Results

A total of 146 interactions of elephants with either treatments or controls were observed (control $n=43$, 2-blend $n=34$; 3-blend $n=69$). In some cases, sex could not be determined from video recordings as the elephants were in herds and individuals could not be clearly observed, although responses

Table 1. Elephant behavioral responses to detection of control, or treated socks with synthetic honey bee alarm pheromone.

Response category	Behavior in response to interacting with control or treatment
0 (no reaction)	No responses, or curiosity but no deterrence, avoidance or stress behavior ¹
1 (Mild concern)	Foot-waving, slight trunk elevation ^{1,2,3}
2 (Substantial concern, repellence)	Ear-flapping, elevated trunk, erect tail, trumpeting, change in travel direction to avoid stimulus ^{1,2,3}
3 (Strong repellence)	Individual alarmed and moves away rapidly to avoid stimulus, or entire herd departs in response to an individual's reaction ^{3,4,5}

¹Wright et al. (2018); ²O'Connell-Rodwell et al. (2007); ³Soltis et al. (2014); ⁴Wright et al. (2023); ⁵King et al. (2007).

could be detected. Positive confirmation of sex was made in 115 cases; of these 99 males and 16 females interacted with the experimental treatments or controls; a considerable proportion of the animals interacting with the treatment showed no response (Fig. 1). Examples of reactions to controls and treatments are shown in the Supporting information.

There was a highly significant treatment effect (Table 2, 3), with the 3-blend treatment producing a significantly higher proportion of 2- and 3-rank responses from elephants than other treatments (single d.f. contrasts: control versus 2- or three-blend: $\chi^2 = 14.54$, $p < 0.0001$; 2-blend versus 3-blend: $\chi^2 = 22.38$, $p < 0.0001$). Strong repellence (3-ranked) reactions only occurred in response to the 3-blend treatment. Herd size was a significant effect in the ordinal logistic regression (Table 2) and GLM for binomial responses (Table 3), with larger herds having a higher probability of individuals reacting to the synthetic alarm pheromones.

Probability of no reaction to the treatments decreased significantly with increasing herd size (Fig. 2). The profile estimates (Fig. 2) show that solo elephants (most typically

adult bulls) have approximately 40% average probability of responding to the pheromone blends. Typical bachelor herds are composed of small numbers of individual bulls (average = 2.16; of 591 groups observed in game counts 1999–2024, only 29 included more than 5 individuals, annual OWNRR game count data, unpubl.). Herds of more than 15 individuals have a predicted probability of at least 75% that at least one individual will respond to the pheromone blend cue (Fig. 2).

Analysis of responses by age group and sex showed that subadult males were slightly more likely to respond to the 3-compound pheromone blend than females, both for ‘substantial concern’ responses (Fig. 3A), and ‘strong repellence’ responses (Fig. 3B). Subadult males had the highest probability of a ‘substantial concern’ response (approximately 40%). Applying a single degree of freedom contrast test for the null hypothesis that subadult individuals had a greater probability to respond than other ages showed a significant effect ($\chi^2 = 146.15$; d.f. = 1; $p < 0.0001$) for ‘strong repellence’ responses, and for ‘substantial concern’ responses

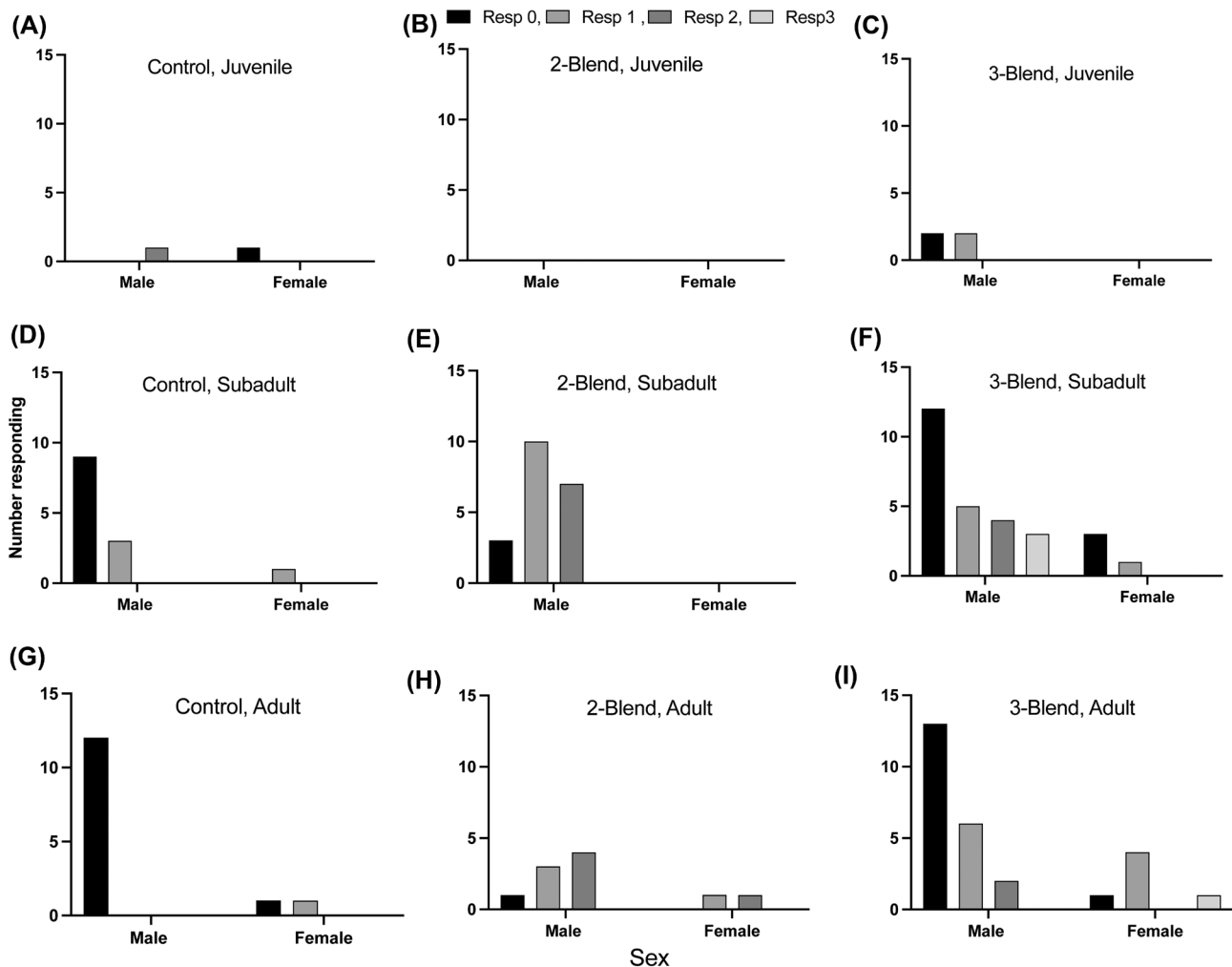


Figure 1. Number of elephants in various demographics responding (Resp 0–3, key above graphs) to synthetic honey bee alarm pheromone blends (2-blend: IAA, 2-heptanone; 3-blend: IAA, 2-heptanone, 2-nonanol), by sex, age and treatment. Table 1 for descriptions of responses.

Table 2. Results of ordinal logistic regression analyses of response strength (ranked 0 through 3) by elephants exposed to synthetic honey bee alarm pheromone blends. (Lack of fit test: -LogLikelihood ratio=61.69; $\chi^2 = 123.38$; $p=0.922$). Bolded sources of variation indicate statistical significance.

Source of variation	χ^2	df	p
Full model	41.13	6	< 0.0001
Treatment	25.93	2	< 0.001
Sex	0.01	1	0.954
Age	1.23	2	0.522
Herd size	8.55	1	0.003

($\chi^2 = 4.94$; $df=1$; $p=0.03$) (Fig. 2). Single degree of freedom contrasts for sex showed that males were significantly more likely to respond than females (substantial concern: $\chi^2 = 7.79$; $df=1$; $p < 0.01$; strong repellence: $\chi^2 = 172.90$; $df=1$; $p < 0.0001$).

Discussion

Consistent with previous work (Wright et al. 2018, 2023), we demonstrated that a proportion of elephants exposed to synthetic honey bee alarm pheromones exhibit a range of reactions, from indifference to heightened vigilance and even repellence. Our results here show that the addition of a third component, 2-nonanol, to the two-component synthetic honey bee alarm pheromone blend originally tested (isoamyl acetate and 2-heptanol, Wright et al. 2018), resulted in increased strength of deterrence reactions, and precipitated retreat responses in a small proportion of elephants encountering the cues. This suggests that 2-nonanol might be a key component inducing bees to launch an actual attack, and this olfactory cue might be learned by elephants as a warning of potential honey bee defensive attack. As previously mentioned, Collins and Blum (1982) showed that 2-nonanol stimulated a substantial and rapid alarm and defensive response in honey bees. It may be that the addition of this compound, and possibly others in the bee blend, might make this formulation more effective than the previous two-component blend containing only isoamyl acetate and heptanol. While previous field work (Wright et al. 2018) quickly produced large numbers of responses to the early two-component blend tested, we had relatively fewer responses in the later work, reported here. This may be due to a more limited exposure of elephants to honey bee attacks in the current study area. The 2018 work was conducted in an area

Table 3. Results of generalized linear model analyses of reactions (binomial, any reaction versus no reaction, logit link function) from elephants exposed to synthetic honey bee alarm pheromone blends. Bolded sources of variation indicate statistical significance.

Source of variation	χ^2	df	p
Whole model test	37.31	6	< 0.0001
Treatment	25.95	2	< 0.0001
Sex	0.26	1	0.326
Age	2.27	2	0.037
Herd size	3.34	1	0.017

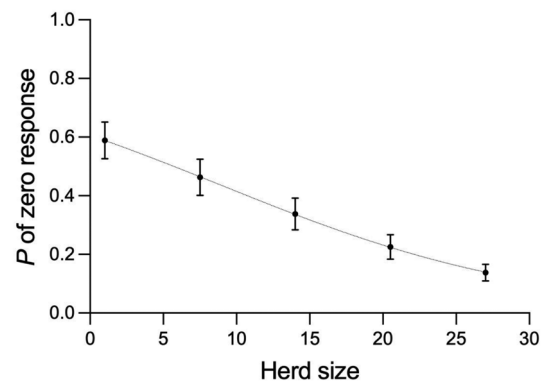


Figure 2. Estimated probability (mean estimate $\pm$ SEM, ordinal logistic regression analysis profiler estimates) of individual elephants demonstrating zero response to synthetic honey bee alarm pheromone treatments in herds of different sizes.

with numerous beehives, part of experimental work to assess the effectiveness of the presence of bee hives for tree protection from elephants (Cook et al. 2023, Cook and Henley 2024). The likelihood that elephants experienced beehives and honey bee attacks during that time in that area were thus relatively high (Wright et al. 2018, Cook and Henley 2024), while beehives were rare on the landscape used for the three-component blend experiments (Transfrontier Africa staff and Wright, unpubl.). This raises the question whether elephants have an inherent aversion to honey bees or if they learn to avoid bees through exposure and negative experiences. It has been demonstrated that elephants learn to avoid negative stimuli and effectively use olfactory cues to distinguish positive and non-rewarding options (Plotnik et al. 2014). Asian elephants rapidly learn to remember and form associations of specific odors with rewards (Arvidsson et al. 2012), and it is assumed that African elephants have similar capacities. Cook and Henley (2024) showed that fear associations with active bee hives were remembered for a considerable time, possibly even years, with elephants still avoiding trees with empty hives. Alternatively, or perhaps additionally, elephants may be inherently averse to these stimuli, given the long cohabitation history elephants and honey bees share in Africa, although their exposure frequency remains unclear. Carefully controlled experiments with captive or semi-habituated herds, where the history of each animal is well known, could provide an option to address these questions. Furthermore, a careful analysis of the literature on the known responses, and possibly neurological processing, of elephants to stress or warning cues may shed light on the nature of the reactions observed. If evidence shows that elephants' reactions to the bee alarm pheromone cues are learned, it might be feasible to use conditioning with live hives as a deterrent, which could later be replaced by the synthetic blend once the elephants have been exposed to bee attacks and learned the alarm cues.

The observed trend for larger herds being much more likely to have at least one individual responding to the alarm blends might be due to the presence of multiple young bulls in breeding herds. These young bulls are typically more

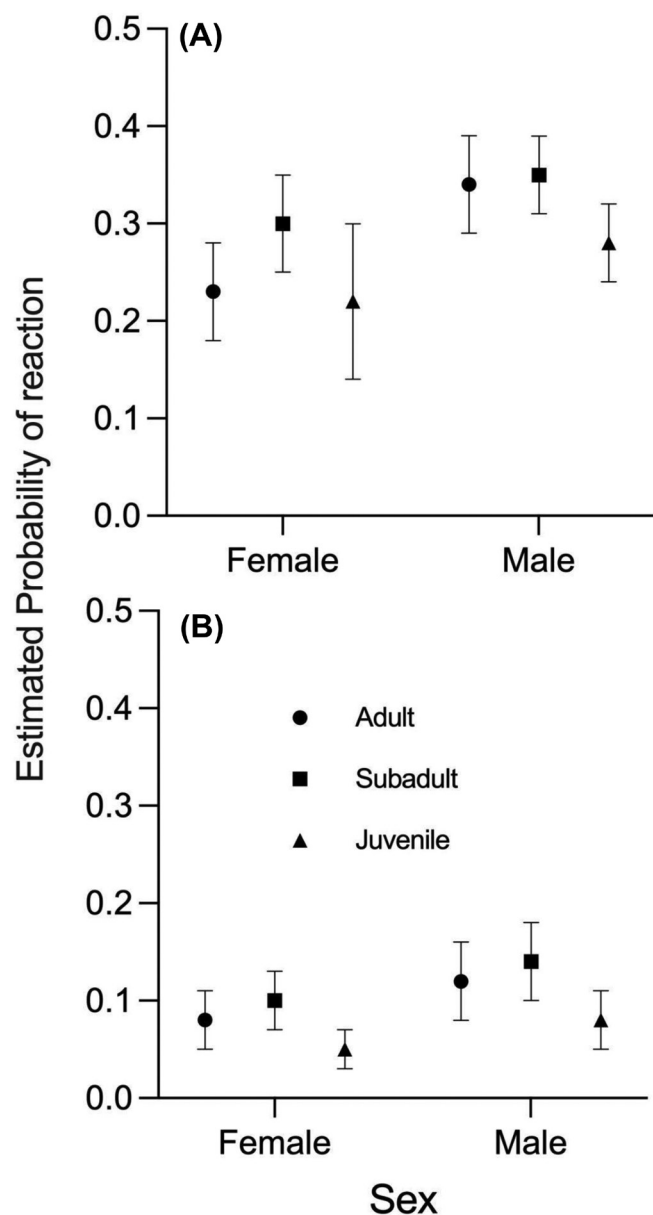


Figure 3. Estimated mean ($\pm$ SEM, ordinal logistic regression analysis profiler estimates) probability of responses of substantial concern (A), and strong repulsion (B), in elephants experiencing synthetic honey bee alarm pheromone blends.

inquisitive and have a higher likelihood of encountering beehives. Our data showed that subadult bulls were indeed slightly more likely to show significant concern or repulsion when experiencing 2- and 3-blend alarm pheromone mixes. Subadult cows were slightly less likely than bulls to react, but more likely to react than adult females, with the caveat that few cows interacted with the treatments, so the probability estimates for this demographic should be viewed with some circumspection. It is perhaps remarkable that so few cows interacted with the treatments, given their protection behavior towards calves, for example. Our observations

suggest that the subadult bulls, rather than cows, were more inclined to explore objects in their surroundings and thus detect the treatments. The demographic composition of a typical breeding herd would, therefore, include multiple individuals of the most reactive age and sex categories. For field applications, a mild deterrence response is ideal. Inducing a panicked mass exodus of elephants could lead to undesirable and dangerous outcomes for the animals. The 3-component blend increased the likelihood of a mild deterrence response.

We have observed reactions of individuals experiencing the blends (Supporting information), followed by immediate responses of others in the herd, which suggest that the elephant first detecting the cue issues a warning signal to the group. It seems plausible that individual elephants might warn others if they anticipate an imminent mass attack by bees that could affect themselves and other herd members. Consequently, a single elephant reacting to the deterrent could cause the entire herd to respond and be deterred from their original path.

Our results show that a relatively simple blend of selected components of the honey bee alarm pheromone chemical assemblage has the potential to produce moderate to strong deterrence responses in African elephants. It seems reasonable to suggest that elephants that have been exposed to negative interactions with honey bees have learned to avoid bees, and their alarm cues. If the use of synthetic alarm pheromone blends is to be considered for elephant behavioral management, it will be necessary to carefully determine to what extent conditioning of herds is required, and how this could be practically achieved in the habitat in question. Distribution of live beehives strategically placed on the landscape to increase exposure of elephants to negative bee interactions would likely provide effective conditioning, similar to the conditioning effect reported by Cook and Henley (2024). Habituation to the warning cues is also a possibility. We did not account for possible effects of repeat exposures, as identifying individuals during waterhole visits was not viable. The fact that herds observed were transient in the area probably reduced the likelihood of multiple exposures, but this remains unquantified. Currently, individual identifications of elephants have not been established, and considering elephants' propensity for habituation this is an area that needs to be explored in future research to establish the long-term potential/efficacy of alarm pheromones as a method of influencing elephant movements.

In addition to improving the potential for managing human–elephant conflict situations, understanding the larger ecological implications of interactions between elephants and honey bees can provide fundamentally important information for conservation management. Investigating the interactions between elephants and wild honey bees in natural settings, for example quantifying the probability of encounter with hives, spatial distribution, and size and age of trees inhabited by bee hives,

could contribute to a better understanding of the role that honey bees may play in protecting mature trees in conservation areas. The expected frequency of elephant–bee interactions would be elucidated, which would contribute to understanding variability in responses to alarm signals among herds and individuals regionally. For example, in the Kruger National Park, there is significant spatial variability in the mortality attributable to elephants, of marula trees (*Sclerocarya birrea*, Anacardiaceae), a favored food tree for elephants (Helm et al. 2009). Marula trees were found more severely impacted by elephants in Sabie thorn thicket ecozone, with some unexplained factors influencing the likelihood of tree utilization and survival. Might honey bees be scarce in thorn thicket, or is this habitat simply more attractive to elephants due to resource availability? Detailed data on the incidence of beehives in different habitats may reveal that some are more hospitable to beehives, and the higher beehive density might provide defense to more trees in those habitats.

While still in the experimental phase, optimized blends of synthetic honey bee alarm pheromones show potential for elephant deterrence. Their inclusion as a passive means of managing elephant behavior offers the potential for a new tool in the integrated management of elephants, which currently includes physical barriers, electric fences, noxious odors, sounds, and direct human intervention. The inclusion of a semiochemical warning stimulus that elephants have evolved with should provide a reduced-stress option for elephant deterrence and reduced human–elephant conflict.

Acknowledgements – The Eva Crane Trust and Ahlborg Zoo are thanked for funding support that sustained this work. OWNRR landowners are thanked for access to study sites, particularly Ingwe Lodge and Naledi Lodge for regular access to hides. Transfrontier Africa staff and volunteers made significant contributions to data collection, and ISCA staff chemists, Carmem Bernardi, Vanessa Soria and chemical ecologist Jesse Saroli are acknowledged for their assistance with SPLAT formulations.

Funding – MGW received grant funding from the Eva Crane Trust. Aalborg Zoo provided funding via Transfrontier Africa.

Permits – This research was conducted under University of Hawaii IACUC, protocol number 17-2561-6.

Author contributions

Mark G. Wright: Conceptualization (equal); Project administration (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Writing – original draft (equal). **Michelle G. Au:** Data curation (equal); Formal analysis (equal); Investigation (equal). **Irene Gatti:** Data curation (equal); Investigation (equal); Project administration (equal). **Juliana Salehi:** Data curation (equal); Investigation (equal). **Craig R. Spencer:** Conceptualization (equal); Investigation (equal). **Paul Allin:** Conceptualization (equal); Investigation (equal). **Agenor Mafra-Neto:** Conceptualization (equal); Investigation (equal).

Transparent peer review

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/wlb3.01419>.

Data availability statement

Data are available from the Dryad Digital Repository: <https://doi.org/10.5061/dryad.pg4f4qs0b> (Wright et al. 2025).

Supporting information

The Supporting information associated with this article is available with the online version.

References

- Allan, S. A., Slessor, K. N., Winston, M. L. and King, G. G. S. 1987. The influence of age and task specialization on the production and perception of honey bee pheromones. – *J. Insect Physiol.* 33: 917–922.
- Angelici, F. M., Gippoliti, S., Herrero, J., Meriggi, A. and Perry, G. 2023. Human–wildlife conflicts: consequences of the erosion of the available habitat, introduction of alien species, and anthropization. – *Front. Ecol. Evol.* 11: 1229398.
- Arvidsson, J., Amundin, M. and Laska, M. 2012. Successful acquisition of an olfactory discrimination test by Asian elephants, *Elephas maximus*. – *Physiol. Behav.* 105: 809–814.
- Boch, R., Shearer, D. A. and Stone, B. C. 1962. Identification of isoamyl acetate as an active component in the sting pheromone of the honey bee. – *Nature* 195: 1018–1020.
- Collins, A. M. and Blum, M. S. 1982. Bioassay of compounds derived from the honey bee sting. – *J. Chem. Ecol.* 8: 463–470.
- Collins, A. M. and Blum, M. S. 1983. Alarm responses caused by newly identified compounds derived from the honey bee sting. – *J. Chem. Ecol.* 9: 57–65.
- Cook, R. M. and Henley, M. D. 2024. Presence and absence of beehives as a management tool for reducing elephant-induced tree mortality. – *Diversity* 16: 577.
- Cook, R. M., Witkowski, E. T. F., Helm, C. V., Henley, M. D. and Parrini, F. 2017. Recent exposure to African elephants after a century of exclusion: rapid accumulation of marula tree impact and mortality, and poor regeneration. – *For. Ecol. Manage.* 401: 107–116.
- Cook, R. M., Witkowski, E. T. F. and Henley, M. D. 2023. Protecting the resource: an assessment of mitigation methods used to protect large trees from African elephant impact in a savanna system. – *Wildl. Biol.* 2025: e01170.
- Ghent, R. L. and Gary, N. E. 1962. A chemical alarm releaser in honey bee stings (*Apis mellifera* L.). – *Psyche* 69: 1–6.
- Goheen, J. R. and Palmer, T. M. 2010. Defensive plant–ants stabilize megaherbivore-driven landscape change in an African savanna. – *Curr. Biol.* 20: 1768–1772.
- Goodyear, S. E. and Schulte, B. A. 2015. Habituation to auditory stimuli by captive African elephants (*Loxodonta africana*). – *Anim. Behav. Cogn.* 2: 292–312.
- Greenwood, D. R., Comeskey, D., Hunt, M. B. and Rasmussen, L. E. L. 2005. Chemical communication: chirality in elephant pheromones. – *Nature* 438: 1097–1098.

- Harding, H. R., Gordon, T. A. C., Eastcott, E., Simpson, S. D. and Radford, A. N. 2019. Causes and consequences of intraspecific variation in animal responses to anthropogenic noise. – *Behav. Ecol.* 30: 1501–1511.
- Helm, C. V., Witkowski, E. T. F., Kruger, L., Hofmeyr, M. and Owen-Smith, N. 2009. Mortality and utilization of *Sclerocarya birrea* subsp. *caffra* between 2001 and 2008 in the Kruger National Park, South Africa. – *S. Afr. J. Bot.* 75: 475–484.
- Howse, P. E., Stevens, I. D. R. and Jones, O. T. 1998. Insect pheromones and their use in pest management. – Chapman & Hall, Springer-Science+Business Media.
- King, L., Pardo, M., Weerathunga, S., Kumara, T. V., Jayasena, N., Soltis, J. and de Silva, S. 2018. Wild Sri Lankan elephants retreat from the sound of disturbed Asian honey bees. – *Curr. Biol.* 28: R64–R65.
- King, L. E., Douglas-Hamilton, I. and Vollrath, F. 2007. African elephants run from the sound of disturbed bees. – *Curr. Biol.* 17: R832–R833.
- King, L. E., Lala, F., Nzumu, H., Mwambingu, E. and Douglas-Hamilton, I. 2017. Beehive fences as a multidimensional conflict-mitigation tool for farmers coexisting with elephants. – *Conserv. Biol.* 31: 743–752.
- Kioko, J., Zink, E., Sawdy, M. and Kiffner, C. 2013. Elephant (*Loxodonta africana*) demography and behaviour in the Taran-gire-Manyara Ecosystem, Tanzania. – *S. Afr. J. Wildl. Res.* 43: 44–51.
- Ma, D., Abrahms, B., Allgeier, J., Newbold, T., Weeks, B. C. and Carter, N. H. 2024. Global expansion of human–wildlife overlap in the 21st century. – *Sci. Adv.* 10: eadp7706.
- Mortimer, B., Walker, J. A., Lolchuragi, D. S., Reinwald, M. and Daballen, D. 2021. Noise matters: elephants show risk-avoidance behaviour in response to human-generated seismic cues. – *Proc. R. Soc. B* 288: 20210774.
- Mumby, H. S. and Plotnik, J. M. 2018. Taking the elephants' perspective: remembering elephant behavior, cognition and ecology in human–elephant conflict mitigation. – *Front. Ecol. Evol.* 6: 122.
- Mutinda, M., Chenge, G., Gakuya, F., Otiende, M., Omondi, P., Kasiki, S., Soriguer, R. C. and Alasaad, S. 2014. Detusking fence-breaker elephants as an approach in human–elephant conflict mitigation. – *PLoS One* 9: e91749.
- Ngama, S., Korte, L., Bindelle, J., Vermeulen, C. and Poulsen, J. R. 2016. How bees deter elephants: beehive trials with forest elephants (*Loxodonta africana cyclotis*) in Gabon. – *PLoS One* 11: e0155690.
- Nouvian, M., Reinhard, J. and Giufra, M. 2016. The defensive response of the honeybee *Apis mellifera*. – *J. Exp. Biol.* 219: 3505–3517.
- O'Connell-Rodwell, C. E., Wood, J. D., Kinzley, C., Rodwell, T. C., Poole, J. H. and Puria, S. 2007. Wild African elephants (*Loxodonta africana*) discriminate between familiar and unfamiliar conspecific seismic alarm calls. *J. Acoust. Soc. Am.* 122: 823–830.
- Palmer, T. M. and Brody, A. K. 2013. Enough is enough: the effects of symbiotic and abundance of herbivory, growth, and reproduction in an African acacia. – *Ecology* 94: 683–691.
- Plotnik, J. M., Shaw, R. C., Brubaker, D. L., Tiller, L. N. and Clayton, N. S. 2014. Thinking with their trunks: elephants use smell but not sound to locate food and exclude nonrewarding alternatives. – *Anim. Behav.* 88: 91–98.
- Soltis, J., King, L. E., Douglas-Hamilton, I., Vollrath, F. and Savage, A. 2014. African elephant alarm calls distinguish between threats from humans and bees. – *PLoS One* 9: e89403.
- Tiller, L. N., Oniba, E., Opira, G., Brennan, E. J., King, L. E., Ndombi, V., Wanjala, D. and Robertson, M. R. 2022. 'Smelly' elephant repellent: assessing the efficacy of a novel olfactory approach to mitigating elephant crop raiding in Uganda and Kenya. – *Diversity* 14: 509.
- Van de Water, A., King, L. E., Arkajak, R., Arkajak, J., van Doormaal, N., Ceccarelli, V., Sluiter, L., Doornwaard, S. M., Praet, V., Owen, D. and Matteson, K. 2020. Beehive fences as a sustainable local solution to human–elephant conflict in Thailand. – *Cons. Sci. Prac* 2: 10.
- Witzgall, P., Kirsch, P. and Cork, A. 2010. Sex pheromones and their impact on pest management. – *J. Chem. Ecol.* 36: 80–100.
- Wood, W. F., Palmer, T. M. and Stanton, M. L. 2002. A comparison of volatiles in mandibular glands from three *Crematogaster* ant symbionts of the whistling thorn acacia. – *Biochem. Syst. Ecol.* 30: 217–222.
- Wright, M. G., Spencer, C. R., Cook, R. M., Henley, M. D., North, W. and Mafra-Neto, A. 2018. African bush elephants response to a honey bee alarm semiochemical blend. – *Curr. Biol.* 28: R778–R780.
- Wright, M. G., Gatti, I., Au, M. G., Salehi, J., Spencer, C. R., Allin, P. and Mafra-Neto, A. 2023. Evaluating formic acid as a behavioral modifier in African savanna elephants. – *Diversity* 15: 1079.
- Wright, M. G., Au, M. G., Gatti, I., Salehi, J., Spencer, C. R., Allin, P. and Mafra-Neto, A. 2025. Data from: Demographic analysis of behavioral reactions of African bush elephants to a synthetic honey bee alarm pheromone blend. – Dryad Digital Repository, <https://doi.org/10.5061/dryad.pg4f4qs0b>.